Quantitative History

量化史学

Margo Anderson | 玛戈·安德森

WHAT IS QUANTITATIVE HISTORY?

什么是量化史学?

Quantitative history is the term for an array of skills and techniques used to apply the methods of statistical data analysis to the study of history. Sometimes also called cliometrics by economic historians, the term was popularized in the 1950s and 1960s as social, political and economic historians called for the development of a 'social science history', adopted methods from the social sciences, and applied them to historical problems. These historians also called for social scientists to historicize their research and consciously examine the temporal nature of the social phenomena they explored. For both types of questions, historians found that they needed to develop new technical skills and data sources. That effort led to an array of activities to promote quantitative history.

量化史学是指将统计数据分析方法应用于历史研究的一系列技能与技术。经济史学家有时也称其为"计量史学"（cliometrics）。这一术语在 20 世纪 50 至 60 年代得到普及，当时社会史、政治史和经济史学者呼吁发展一种"社会科学史学"，借用社会科学的方法并将其运用于历史问题。同时，这些历史学家也主张社会科学家应当使自己的研究历史化，自觉考察其所研究的社会现象所具有的时间性。无论面对哪一类问题，历史学家都发现自己需要发展新的技术技能与数据来源。由此形成了一系列推动量化史学发展的活动。

Classical historical research methodology relies upon textual records, archival research and the narrative as a form of historical writing. The historian describes and explains particular phenomena and events, be they large epic analyses of the rise and fall of empires and nations, or the intimate biographical detail of an individual life. Quantitative history is animated by similar goals but takes as its subject the aggregate historical patterns of multiple events or phenomena. Such a standpoint creates a different set of issues for analysis. A classic historical analysis, for example, may treat a presidential election as a single event. Quantitative historians consider a particular presidential election as one element in the universe of all presidential elections and are interested in patterns which characterize the universe or several units within it. The life-course patterns of one household or family may be conceived as one element in the aggregate patterns of family history for a nation, region, social class or ethnic group. Repeated phenomena from the past that leave written records, which read one at a time would be insignificant, are particularly useful if they can be aggregated, organized, converted to an electronic database and analyzed for statistical patterns. Thus records such as census schedules, vote tallies, vital (e.g., birth, death and marriage) records; or the ledgers of business sales, ship crossings, or slave sales; or crime reports permit the historian to retrieve the pattern of social, political, and economic activity in the past and reveal the aggregate context and structures of history.

传统的历史研究方法依赖文本记录、档案研究以及作为历史写作形式的叙事。历史学家描述并解释特定的现象与事件，无论是关于帝国和国家兴衰的宏大分析，还是对个人生命细节的细致传记。量化史学具有相似的目标，但它把多个事件或现象的总体性历史模式作为研究对象。这样的立场带来了另一组分析问题。例如，传统历史分析可能把一次总统选举视为一个单独事件；量化史学家则把某一次总统选举看作全部总统选举这一总体中的一个要素，并关注能够刻画整个总体或其中若干单位的模式。一个家庭或家户的生命历程模式，也可以被理解为一个国家、地区、社会阶层或族群家庭史总体模式中的一个组成部分。过去那些反复出现并留下书面记录的现象，如

果逐条阅读也许显得微不足道，但若能够加以汇总、整理、转换为电子数据库并进行统计模式分析，就尤其有用。因此，人口普查表、选票统计、生命事件记录（如出生、死亡和婚姻记录），商业销售、船只航行或奴隶买卖账簿，以及犯罪报告等资料，都使历史学家能够重建过去社会、政治和经济活动的模式，并揭示历史的总体背景与结构。

The standpoint of quantitative history also required a new set of skills and techniques for historians. Most importantly, they had to incorporate the concept of the data set and data matrix into their practice. Floud (1972: 17) defined the data set as ‘a coherent selection of data from the whole range of historical data available to the historian, and it is selected because it relates closely to the questions that the historian wishes to consider.’ The myriad instances of a phenomenon—for example, all United States presidential elections—form the cases of the data set. The pieces of information collected about the cases—for example, the candidates running, the year of the election or the vote totals—become the variable characteristics of the data set, that is, the varying characteristics of any particular case. The historian arranges the data in tabular form, that is, in a matrix of rows and columns, ‘consisting of a number of rows, which will normally represent cases, and a number of columns, which will normally represent variables’ (Floud, 1972: 18). The creation of quantitative data sets thus required the historian to carefully compile consistent information about the phenomenon to be investigated, and prepare the data in tabular form. Historians then were prepared to apply the techniques of statistical data analysis to the data set to answer the research question posed.

量化史学的这种研究立场也要求历史学家具备一套新的技能与技术。最重要的是，他们必须把“数据集”和“数据矩阵”的概念纳入研究实践。Floud（1972：17）把数据集定义为：“从历史学家能够获得的全部历史数据中有系统地选择出来的一组数据之所以被选中，是因为它与历史学家希望考察的问题密切相关。”一个现象的大量实例——例如美国历次总统选举——构成数据集中的“案例”。围绕这些案例收集的信息——例如参选候选人、选举年份或得票总数——则成为数据集中的变量特征，也就是不同案例之间会发生变化的特征。历史学家把数据整理为表格形式，即行列矩阵：“由若干行组成，行通常代表案例；由若干列组成，列通常代表变量”（Floud，1972：18）。因此，建立量化数据集要求历史学家细致地汇集关于研究现象的一致性信息，并把数据整理成表格形式。在此基础上，历史学家才能运用统计数据分析技术来回答提出的研究问题。

In short, to make effective use of quantitative evidence and statistical techniques for historical analysis, practitioners had to integrate the rapidly developing skills of the social sciences, including sampling, statistical data analysis and data archiving into their historical work. That task led to the development of new training programs in quantitative methods for historians, to the creation of new academic journals and textbooks, and to the creation of data archives to support the research.

简言之，为了在历史分析中有效使用量化证据和统计技术，研究者必须把社会科学迅速发展的技能——包括抽样、统计数据分析和数据存档——整合进历史研究。由此推动了面向历史学家的量化方法培训项目、新的学术期刊和教材的创办，也促成了为研究提供支持的数据档案的建立。

Early Efforts

早期探索

Historians had made use of quantitative evidence prior to the 1950s, particularly in the fields of economic and social history. The Annales school in France pointed the way in the pre-World War II period. The rapid growth and expansion of the United States had long required American historians to consider quantitative issues in their study of the growth of the American economy, population and mass democracy. Thus, for example, Frederick Jackson Turner's classic 1893 essay on ‘The Significance of the Frontier in

American History' was largely based on a reading and interpretation of the results of the 1890 population census.

在 20 世纪 50 年代以前，历史学家已经使用量化证据，尤其是在经济史与社会史领域。法国的年鉴学派在第二次世界大战前已经指明了这一方向。美国的迅速增长与扩张也长期促使美国历史学家在研究美国经济、人口和大众民主的发展时考虑量化问题。例如，弗雷德里克·杰克逊·特纳 1893 年的经典论文《美国历史中边疆的意义》，在很大程度上就是建立在对 1890 年人口普查结果的阅读和解释之上的。

But true 'data analysis' in the current sense had to await the growth of the social and statistical sciences in the first half of the twentieth century, and the diffusion to universities in the 1950s of the capacity for machine tabulation of numerical records, and then of mainframe computing in the 1960s. One can see the emerging field exemplified in seminal studies in the late 1950s and early 1960s. In 1959, for example, Merle Curti and his colleagues at the University of Wisconsin published The Making of an American Community: A Case Study of Democracy in a Frontier County. Curti et al. (1959) explored Turner's thesis with an in-depth look at the mid-nineteenth century history of Trempeleau County, Wisconsin, including its records of newspapers, diaries, private papers and county histories. But they also added data analysis of the employment patterns derived from the individual-level federal census manuscripts for the censuses from 1850 through 1880.

不过，现代意义上的真正“数据分析”，必须等到 20 世纪上半叶社会科学与统计科学的发展，以及 20 世纪 50 年代数字记录的机器制表能力进入大学、60 年代大型主机计算普及之后才成为可能。20 世纪 50 年代末和 60 年代初的一些奠基性研究体现了这一新兴领域。比如，1959 年，威斯康星大学的 Merle Curti 及其同事出版了《一个美国社区的形成：边疆县民主的个案研究》。Curti 等人（1959）通过深入考察 19 世纪中叶威斯康星州 Trempeleau County 的历史来检验特纳的论题，使用了当地报纸、日记、私人文书和县史等资料；与此同时，他们还利用 1850 至 1880 年联邦人口普查的个人层级手稿数据，对就业模式进行了数据分析。

Similarly, the 'new' economic historians of the 1950s challenged the conventional wisdom of the day on several key issues in economic history. One debate centered on the 'necessity' of the US Civil War. Historians at the time argued that the war had been 'unnecessary' since the institution of race-based slavery would collapse under the weight of its unprofitability. In contrast, economic historians employed economic theory and data on output of southern agriculture to argue that the southern agricultural economy could have survived profitably into the twentieth century using slave labor (Conrad and Meyer, 1958). Robert Fogel challenged the conventional wisdom on the centrality of railroads for the industrial development of the United States. Making use of economic theory, carefully compiled data series, and the logic of the counterfactual, Fogel argued that canals would have also succeeded as a transportation system underpinning nineteenth-century American industrial development (1964).

同样，20 世纪 50 年代的“新”经济史学家也对当时经济史研究中的若干重要传统观点提出挑战。其中一场争论围绕美国内战是否“必要”展开。当时一些历史学家认为战争其实“没有必要”，因为以种族为基础的奴隶制度会由于自身缺乏盈利能力而最终崩溃。与此相反，经济史学家利用经济理论和美国南部农业产出数据，论证南部农业经济即使依赖奴隶劳动，也可能以盈利方式延续到 20 世纪（Conrad and Meyer，1958）。Robert Fogel 则挑战了铁路对于美国工业发展具有核心作用这一传统认识。他运用经济理论、精心整理的数据序列以及反事实推理，提出运河同样可能成功地充当支撑 19 世纪美国工业发展的交通体系（1964）。

'New political historians' such as Lee Benson, Allan Bogue, Richard P. McCormick, and political scientists with historical interests, such as Warren Miller and Walter Dean Burnham, translated the emerging techniques of political scientists analyzing contemporary election results and voter surveys to historical questions, and opened up dramatic new insights into American political history.1 The new

political historians identified the parameters of party systems, developed the theory of the critical election, and argued that underlying structures of electoral politics were accessible through historical analysis of voter turnout and election results. In 1964 in England, demographers and historians founded the Cambridge Group for the History of Population and Social Structure and began a forty-year project to retrieve, assemble and reconstruct 400 years of the family history of Britain.2

Lee Benson、Allan Bogue、Richard P. McCormick 等“新政治史学家”，以及 Warren Miller、Walter Dean Burnham 等具有历史兴趣的政治学家，把政治学家分析当代选举结果和选民调查的新兴技术转用于历史问题，从而为美国政治史带来了重要的新认识。[1] 新政治史学家识别了政党体系的基本参数，发展了“关键选举”理论，并认为可以通过对选民投票率和选举结果的历史分析来把握选举政治的深层结构。1964 年，英国的人口学家和历史学家成立了剑桥人口与社会结构史研究组（Cambridge Group for the History of Population and Social Structure），并启动了一个持续四十年的项目，旨在搜集、汇编并重建英国四百年的家庭史。[2]

The new possibilities of quantitative history fit well with other trends within the discipline of history, particularly with the growth of social history and calls for what Jesse Lemisch (1967) called ‘history from the bottom up’—that is, for historians to treat the lives of ordinary people, to complement the study of elites. By the mid-1960s, the interest in the new techniques led the American Historical Association to recognize that ‘quantification in history’ would require new skills and institutions within the historical profession. The AHA created a Quantitative Data Committee to consider the issues. Summer institutes and classes in quantitative methods for historians were held in 1965, 1967 and 1973 at the University of Michigan, Cornell University and Harvard University respectively. In 1968, the Inter-university Consortium for Political Research at the University of Michigan began offering a four-week course in quantitative historical analysis as part of its summer program in quantitative methods. The course continues to be offered each summer. At the Newberry Library in Chicago, from 1971 to 1982 Richard Jensen spearheaded a summer program in quantitative methods for historians. By the early 1980s, about 40 percent of history graduate programs offered training in quantitative history as part of the graduate curriculum (Bogue, 1983: 220ff.).3

量化史学带来的新可能与历史学内部的其他趋势十分契合，尤其是社会史的兴起，以及 Jesse Lemisch（1967）所谓“自下而上的历史”（history from the bottom up）的倡议——也就是要求历史学家在研究精英的同时，把普通人的生活纳入历史研究。到 20 世纪 60 年代中期，对这些新技术的兴趣促使美国历史学会认识到，“历史中的量化”需要历史学专业内部发展新的技能和制度。美国历史学会设立了量化数据委员会来讨论相关问题。1965 年、1967 年和 1973 年，密歇根大学、康奈尔大学和哈佛大学分别举办了面向历史学家的量化方法暑期研习班和课程。1968 年，密歇根大学的大学间政治研究联盟开始在其量化方法暑期项目中开设为期四周的量化历史分析课程，该课程此后每年夏季持续开设。1971 至 1982 年间，Richard Jensen 在芝加哥纽伯里图书馆主持了面向历史学家的量化方法暑期项目。到 20 世纪 80 年代初，大约 40%的历史学研究生项目已在研究生培养课程中提供量化史学训练（Bogue，1983：220ff.）。[3]

Additional institutional infrastructure of quantitative history can also be dated to the 1960s. New journals, textbooks, and edited collections also promoted the growth of quantitative history. The Historical Methods Newsletter, for example, began publishing in 1967, and was renamed Historical Methods in 1978. The Journal of Interdisciplinarity History began publication in 1970. The Social Science History Association (SSHA) was founded in 1974 and the first issue of its journal, Social Science History, appeared in 1976. SSHA became the professional venue for bringing together historians who consciously adopted theory and methodology from the social sciences and social scientists doing historical work. The cross-fertilization has continued, and, as noted below, many of the innovations in quantitative history have been developed by

scholars with formal training in the social sciences and appointments in departments of economics, demography, sociology, anthropology, geography and political science.

量化史学的其他制度基础也可以追溯到 20 世纪 60 年代。新的期刊、教材和论文集同样推动了量化史学的发展。例如，《Historical Methods Newsletter》创刊于 1967 年，1978 年更名为《Historical Methods》。《Journal of Interdisciplinarity History》于 1970 年创刊。社会科学史协会（Social Science History Association，SSHA）成立于 1974 年，其期刊《Social Science History》于 1976 年出版第一期。SSHA 成为一个专业平台，把有意识地采用社会科学理论与方法的历史学家，以及从事历史研究的社会科学家汇聚起来。这种交叉融合一直延续至今；如下文所述，量化史学中的许多创新都由受过正规社会科学训练、任职于经济学、人口学、社会学、人类学、地理学和政治学等系科的学者发展出来。

Textbooks in quantitative history began to appear in the early 1970s, and many have been published since.4 Numerous edited volumes introduced the new field and techniques to professional and student audiences.5 Finally, researchers created data archives. In the United States, the Inter-university Consortium for Political Research (ICPR) was founded in 1962 primarily by political scientists. Renamed the Inter-university Consortium for Political and Social Research (ICPSR) in 1975, the Consortium has also pioneered in the creation and preservation of historical data collections. The United States National Archives and Records Administration (NARA) created an electronic records preservation program in the early 1970s for federal government data that was ‘born digital’ (Ambacher, 2003; Adams, 1995, forthcoming; Fishbein, 1973). Similar work began in Britain with the founding of the UK Data Archive in 1967.6

量化史学教材在 20 世纪 70 年代初开始出现，此后出版了许多相关著作。[4] 大量论文集也向专业研究者和学生介绍这一新领域及其技术。[5] 最后，研究者还建立了数据档案。在美国，大学间政治研究联盟（Inter-university Consortium for Political Research，ICPR）主要由政治学家于 1962 年创建。1975 年，该联盟更名为大学间政治与社会研究联盟（Inter-university Consortium for Political and Social Research，ICPSR），并在历史数据集的建设和保存方面发挥了开创性作用。美国国家档案和记录管理局（NARA）在 20 世纪 70 年代初建立电子记录保存项目，用于保存联邦政府“原生数字”（born digital）数据（Ambacher，2003；Adams，1995，forthcoming；Fishbein，1973）。英国也开展了类似工作，1967 年成立了英国数据档案馆（UK Data Archive）。[6]

Thus by 1980, historians had taken major steps to establish the institutional structures necessary to integrate quantitative history into larger historical practice. That infrastructure has, if you will, both matured and faced challenges in the generation of work since, and in many ways quantitative history is still a work in progress. Nevertheless, it is possible to identify the types of questions quantitative history was intended to and has been able to address; the major types of data sets that have developed and the key characteristics of historical data sets; and the most commonly used techniques within the field. That background in turn provides the framework for a review of a number of methodological issues historians uniquely face, for a review of the achievements of quantitative history, and for a discussion of emerging issues.

因此，到 1980 年，历史学家已经在建立必要的制度结构方面迈出了重要步伐，使量化史学能够融入更广泛的历史研究实践。此后的一个世代中，这套基础设施一方面逐渐成熟，另一方面也面临挑战；从许多方面看，量化史学至今仍是一项持续发展的工作。不过，我们仍然能够辨认出量化史学最初旨在回答并且已经能够回答的问题类型、已经发展出来的主要数据集类型及历史数据集的关键特征，以及这一领域最常用的技术。这些背景又为进一步讨论历史学家所面对的一系列独特方法论问题、评估量化史学的成就，以及讨论新兴议题提供了框架。

Questions, Data and Issues in Creating Historical Data Sets

创建历史数据集：问题、数据与议题

Quantitative history has been most successful in addressing big questions about long-term historical patterns of change. Practitioners have achieved important results by assembling substantial amounts of numeric or countable information, and organizing it into tabular data matrices for statistical analysis. The first generation of studies focused especially on the history of the family and social structure, trends in economic growth and change, patterns of electoral behavior and voter participation, or the record of inter-generational social mobility and living standards. More recently, the examples have proliferated. Historians of crime and the criminal justice system, for example, have retrieved court and newspaper records to examine the long-term patterns of crime and violence in the past. Historians of the family have examined patterns of inheritance and the inter-generational transfer of wealth. The emerging work of 'anthropometric' history—the study of living standards and well-being in the past using measures of height, weight, stature and disease in the past—has cast an even wider net, aiming to evaluate comparative living standards over centuries and ultimately millennia.7

量化史学在处理有关长期历史变化模式的重大问题方面最为成功。研究者通过汇集大量数字化或可计数的信息，并把它们组织成表格数据矩阵进行统计分析，取得了重要成果。第一代研究尤其关注家庭与社会结构史、经济增长与变化趋势、选举行为和选民参与模式，以及代际社会流动与生活水平的历史记录。近年来，相关研究案例不断扩展。例如，犯罪史和刑事司法史研究者从法院记录和报纸中提取资料，以考察过去犯罪与暴力的长期模式；家庭史研究者分析继承模式和财富的代际转移。新兴的"人体测量史"（anthropometric history）——即利用过去人口的身高、体重、体格和疾病等指标研究历史上的生活水平与福祉——把研究范围扩展得更广，目标是比较跨越数百年乃至数千年的生活水平。[7]

Making such studies possible was an explosive growth in the data sets informing quantitative history. Quantitative history, like other branches of the social sciences, requires what was once called 'machine-readable' (and are now known as 'electronic') data for analysis. Though there are some examples of large-scale data analysis undertaken by manual systems of tabulation and statistical analysis, most notably the nineteenth-century tabulations of census or vital registration records, social science data in the modern sense required the development of machine tabulation devices, counter sorters, and other mechanized calculators. The first system was the Hollerith system of punch-card tabulation used for the 1890 American population census; the social and statistical sciences grew with the new machinery. By the 1940s, social scientists had developed rules and procedures for collecting quantitative data to make best use of machine tabulation and analysis. These conventions included the fixed format data matrix, the classification of variables into nominal, ordinal, interval and ratio variables, the organization of questionnaires and survey forms to facilitate conversion to punch-cards for analysis, and coding systems such as the Likert scale. Quantitative historians inherited these practices and adapted this existing technology and set of conventions to their historical project. They soon recognized that they had to solve major new methodological and logistical problems before the potential for quantitative history could be achieved.

使这些研究成为可能的，是支撑量化史学的数据集出现了爆炸式增长。与社会科学的其他分支一样，量化史学分析需要过去所谓"机器可读"（machine-readable）、现在通常称为"电子"数据的资料。尽管历史上也存在利用人工制表与统计分析系统进行大规模数据分析的例子，最典型的是 19 世纪对人口普查或生命登记记录进行的汇总，但现代意义上的社会科学数据仍然依赖机器制表装置、计数分类器及其他机械计算设备的发展。最早的系统是 1890 年美国人口普查所采用的 Hollerith 穿孔卡制表系统；社会科学与统计科学也随着这些新机器而发展。到 20 世纪 40 年代，社会科学家已经形成一套收集量化数据的规则和程序，以便充分利用机器制表与分析。这些惯例包括固定格式的数据矩阵，把变量划分为名义变量、顺序变量、区间变量和比率变量，设计问卷和调查表以

便转换为穿孔卡进行分析，以及使用 Likert 量表等编码系统。量化史学家继承了这些实践，并把既有技术和惯例调整后用于历史研究。他们很快意识到，要真正实现量化史学的潜力，还必须解决一系列新的重大方法论与后勤问题。

The first problem derives from the larger evidentiary issue faced by all historians, namely, that historical analysis must rely on the extant record of the past. Historians are at the mercy of their subjects' penchant and capacity for preservation. And before 1890, that is, for most of the historical record of human history, no preserved data were 'machine-readable'. Thus all potential historical data had to be created from surviving, usually text-based, records and converted to machine-readable or electronic format. Even records collected in the twentieth century and informed by the conventions of the emerging social sciences frequently no longer exist in machine-readable format. Thus, the United States Census Bureau, for example, preserved the original paper census questionnaires from the eighteenth century forward. But census officials did not retain the punch-cards they used to tabulate the censuses from 1890 to 1960. These cards were destroyed once the results of the census appeared in published form. Thus historians interested in reanalyzing the microdata from past censuses faced creating, or recreating, the machine-readable records.

第一个问题源自所有历史学家都会面对的更普遍的证据问题：历史分析必须依靠过去遗留下来的现存记录。历史学家在很大程度上受制于研究对象保存资料的意愿和能力。而在 1890 年以前——也就是说，在人类历史绝大多数时期留下的历史记录中——没有任何保存下来的数据是“机器可读”的。因此，所有潜在的历史数据都必须从幸存下来的、通常以文本为基础的记录中重新创建，并转换为机器可读或电子格式。即使是 20 世纪收集、并受到新兴社会科学规范影响的记录，也往往已经不再以机器可读格式存在。例如，美国人口普查局保存了自 18 世纪以来的原始纸质人口普查问卷，却没有保留 1890 至 1960 年人口普查制表时使用的穿孔卡；一旦普查结果正式出版，这些卡片就被销毁了。因此，希望重新分析过去人口普查微观数据的历史学家必须重新创建或恢复这些机器可读记录。

Quantitative historians faced additional major methodological problems resulting from the recalcitrance of the existing archival historical records. All historians face the problems of missing data, and the difficulties of interpreting illegible, damaged, incomplete or destroyed records. For quantitative historians, though, aiming to translate the archival record to a data matrix for statistical analysis, these questions of data quality are particularly difficult. Cases and variables for a data matrix require precise conceptual and operational definitions, as do the allowable entries for particular cell values within the matrix, since the goal of statistical analysis is to assess extent, central tendency and dispersion of any particular characteristic. What does one do if the records for a year or period of years are missing? How does one handle illegible entries in the records of a company's finances? How does one know if the probate records found in a county archive are complete? Historians have had to confront the requirements for case and variable definition, classification and coding in building a data set. The solutions to these problems emerged with the overall field. The journal, Historical Methods, in particular, became the venue for identifying, debating and proposing methodological solutions to these issues.

现存历史档案本身的“顽固性”还给量化史学家带来了其他重大的方法论问题。所有历史学家都会遇到数据缺失，以及记录字迹不清、受损、不完整或被毁等解释困难。然而，对于希望把档案记录转化为数据矩阵并进行统计分析的量化史学家而言，数据质量问题尤其棘手。数据矩阵中的案例和变量需要精确的概念定义与操作定义，矩阵中各单元格所允许的取值也必须清楚规定，因为统计分析的目标正是评估某一特征的范围、集中趋势和离散程度。如果某一年或若干年的记录缺失，应如何处理？公司财务记录中无法辨认的条目怎么办？如何判断某个县档案馆保存的遗嘱认证记录是否完整？在建立数据集时，历史学家必须正视案例和变量的定义、分类与编码要

求。随着整个领域的发展，这些问题逐渐形成相应解决方案。尤其是《Historical Methods》期刊，成为识别、讨论并提出这些方法论问题解决方案的重要平台。

A related issue is the set of rules for extracting the information from a text-based evidentiary source to create a data set. Historical archives frequently contain text-based records that lend themselves to data set construction, but require considerable conceptual work before they can be manipulated statistically. Historians have made use of sales invoices, wills, parish registers and case files of charity or social welfare agencies, for example, and have had to create the cases and variables from the extant texts.

与此相关的另一个问题，是如何制定规则，从以文本为基础的证据来源中提取信息并建立数据集。历史档案中经常包含适合构建数据集的文本记录，但在能够进行统计处理之前，需要完成大量概念性工作。例如，历史学家曾利用销售发票、遗嘱、教区登记簿以及慈善机构或社会福利机构的个案档案，并必须从这些现存文本中自行构造案例与变量。

Historians have had to solve these methodological questions as they select the evidence to be analyzed and create the code-book for the data set. Whether one is analyzing existing tabular data from the past—for example, the records of imports and exports of a nation over a period of years, or the published results of a census—or whether one is creating a data set from text-based sources, the historian needs to define the case or unit of analysis, define the characteristics or variables to be selected to characterize the cases within the data set, and define the coding system used to organize the source information for the data set. Several examples of the issues involved best illustrate the work of quantitative historians.

历史学家在选择要分析的证据并为数据集建立编码手册时，必须解决这些方法论问题。无论是分析过去已有的表格数据——例如某国若干年间的进出口记录或已公布的人口普查结果——还是从文本资料中创建数据集，历史学家都需要界定案例或分析单位，界定用于描述数据集中案例的特征或变量，并界定整理原始资料信息时所采用的编码体系。下面几个例子最能说明量化史学家的具体工作。

Compilation and Analysis of Previously Published Data

既有出版数据的汇编与分析

The most accessible sources for quantitative historians were data that were already published in tabular format. The first generation of quantitative historians in particular compiled data sets from existing, usually aggregated, published data sources—for example, tabulated census results, election results, government reports of tax collections, imports and exports, and data from trade publications. Assembled into time series, such data permitted researchers to undertake basic analyses of historical trends and use regression models to correlate the determinants of change. For example, Walter Dean Burnham's 950+-page study of count-level presidential election results, published in 1955, included a compilation of results from state archives and newspaper sources, and a discussion of the methodological issues he faced in compiling the data. Combined with denominator data from census results that allowed the researchers to measure turnout, the new data set permitted Burnham and his colleagues to begin the analysis of historical election analysis (Burnham, 1955). In similar ways, economic historians made particularly good use of the data compiled in statistical abstracts, such as the Statistical Abstract of the United States, published annually since 1878.

量化史学家最容易使用的资料，是已经以表格形式公布的数据。第一代量化史学家尤其常从现有的、通常已经汇总的公开数据来源中编制数据集，例如人口普查汇总结果、选举结果、政府关于税收、进出口的报告，以及行业出版物中的数据。把这些数据整理为时间序列之后，研究者就可以对历史趋势进行基础分析，并利用回归模型考察变化的决定因素之间的关联。例如，Walter Dean Burnham 于 1955 年出版了一部超过 950 页的总统选举县

级结果研究，其中汇编了州档案和报纸来源中的结果，也讨论了他在整理数据过程中遇到的方法论问题。再结合人口普查结果所提供的分母数据，研究者就可以计算投票率；这一新的数据集使 Burnham 及其同事得以开始对历史选举展开分析（Burnham，1955）。同样，经济史学家也特别善于利用统计摘要中汇编的数据，例如自 1878 年以来每年出版的《Statistical Abstract of the United States》。

Converting Tabular Data in Manuscript Form to Electronic Format
把手稿形式的表格数据转换为电子格式

A second source of quantitative data were archived tabular records in text-based format, probably best illustrated by individual-level census manuscript schedules. See Figures 14.1 and 14.2,8 a facsimile of the 1950 US Census population schedule.

量化数据的第二类来源，是以文本形式保存的档案表格记录，其中最典型的是个人层级的人口普查手稿表。图 14.1 和图 14.2[8]展示了 1950 年美国人口普查人口表的摹本。

For the United States, such original census responses are available for all the federal censuses except 1890, and are available for public use through 1930.9 The schedule is already in a matrix format, with rows of cases and columns of variables. The original difficulty with using these records is their volume. With one record per person for the censuses of 1850 and later, data set creation for a large portion of the population was beyond the capacities of an individual researcher. The first generation of quantitative historians resolved this problem by sampling, and usually by organizing a research project of a particular locale. The historical social mobility studies were designed as community studies to solve the problem of the volume of data.

在美国，除 1890 年外，历次联邦人口普查的这类原始答卷都仍有保存；截至原文写作时，可供公众使用的资料开放到 1930 年人口普查。[9] 这些表格本身已经具有矩阵结构：行代表案例，列代表变量。使用这些记录最初面临的困难是数据量过大。自 1850 年及此后的普查起，每个人都有一条记录，因此，一个研究者很难独立完成大规模人口数据集的建立。第一代量化史学家通过抽样来解决这个问题，通常把研究组织为针对某一特定地方的项目。历史社会流动研究常被设计为社区研究，正是为了克服数据规模问题。

Later generations of quantitative historians have by and large solved the problem of volume through collaboration and by building historical public-use microdata samples, or PUMS files. Starting with the 1900 census, historians proposed to create historical PUMS files that would be similar to the contemporary PUMS files that the Census Bureau has created since 1970. In the late 1980s, researchers at the University of Minnesota, initially led by Russell Menard, Steven Ruggles and Robert McCaa, began systematic retrieval of the historical census data from the United States, and more recently from other nations. The Integrated Public Use Microdata Sample (IPUMS) Project and the International IPUMS project have created microdata samples for the United States from all the censuses from 1850 to 2000, and are now collecting such data for many nations of the world. The data are easily downloadable from the web. The researchers have also built the code-books, technical support materials, and research bibliography necessary for the user to understand the context of the questions and responses to the census.10

后来的量化史学家在很大程度上通过合作，以及建立历史公共使用微观数据样本（PUMS 文件），解决了数据量的问题。从 1900 年人口普查开始，历史学家提出建立历史 PUMS 文件，使其类似于美国人口普查局自 1970 年以来制作的当代 PUMS 文件。20 世纪 80 年代末，明尼苏达大学的研究者最初在 Russell Menard、Steven Ruggles 和 Robert McCaa 的领导下，开始系统提取美国历史人口普查数据，后来又扩展到其他国家。综合公共使用微观数据样本项目（IPUMS）和国际 IPUMS 项目已经为美国 1850 至 2000 年间的历次人口普查建立了微观

数据样本，目前还在收集世界许多国家的相关数据。这些数据可以方便地从网上下载。研究者还建立了编码手册、技术支持材料和研究书目，使使用者能够理解人口普查问题与回答的具体语境。[10]

Creating Tabular Data from Text-Based Records

从文本记录创建表格数据

The most time-consuming type of data set creation is the conversion of text-based records to matrix format. For existing tabular data, whether in manuscript or published form, the basic framework of the matrix is given in the original source. For text-based records with no tabular structure, it is up to the researcher to create the code-book, and thus all the variable definitions and coding rules. Figure 14.3,11 an illustration of a record of a slave sale in antebellum America, illustrates the issues.12

最耗时的数据集创建方式，是把文本记录转换为矩阵格式。对于已有的表格数据，无论是手稿还是出版物形式，矩阵的基本框架已经由原始资料给出；但对于没有表格结构的文本记录，则需要研究者自行编制编码手册，也就是定义所有变量及其编码规则。图 14.3[11]展示了一份美国内战前时期的奴隶买卖记录，它很好地说明了这些问题。[12]

There are thousands of such records in newspapers, private collections and archives, and, if marshaled for analysis, provide detailed, if somewhat gruesome, evidence of this chapter in American economic history. Robert Fogel and Stanley Engerman compiled such records for their study, Time on the Cross (1974) from the New Orleans Slave Market. ICPSR Study 7423 contains the data and code-book for the New Orleans Slave Sale Sample.13

报纸、私人收藏和档案馆中存在成千上万份这样的记录；如果把它们汇集起来进行分析，就能为美国经济史的这一章节提供细致、尽管令人不适的证据。Robert Fogel 与 Stanley Engerman 在《Time on the Cross》（1974）的研究中，从新奥尔良奴隶市场汇编了此类记录。ICPSR Study 7423 保存了“新奥尔良奴隶买卖样本”（New Orleans Slave Sale Sample）的数据和编码手册。[13]

Figure 14.1 Facsimile of 1950 Census Schedule for Orange City, Iowa (left side of page)

图 14.1 1950 年爱荷华州奥兰治城人口普查表摹本（页面左侧）

Original figure / 原图

The uploaded Markdown contains this figure as a remote image link.
上传的 Markdown 中该图以远程图片链接形式存在。

Open original image / 打开原图

For their sample, Fogel and Engerman converted the text-based records into cases and variables and codes, making decisions on unit of analysis (the slave), sampling (2.5 percent or 5 percent, depending on the year of sale), number of variables (46), and codes. Each decision extracted a piece of information from the original text-based records, and had implications for ultimate analysis. The final data set contained 5009 records, and included information on the characteristics of the slave (e.g., age, sex, occupation, color), the terms of the sale (e.g., the date, price, whether paid in cash, the number of slaves sold together), and information on the buyer and seller. Fogel and Engerman used the data to analyze the inter-state slave trade, and to address questions about the economic viability of the slave economy (Fogel and Engerman, 1974).

在他们的样本中，Fogel 和 Engerman 把文本记录转换为案例、变量和编码，并对分析单位（奴隶本人）、抽样比例（视出售年份而定，为 2.5%或 5%）、变量数量（46 个）以及编码方式作出决定。每一个决定都从原始文本记录中提取了一部分信息，并会影响最终分析。最终数据集包含 5009 条记录，其中包括奴隶的特征信息（如年龄、性别、职业、肤色）、交易条件（如日期、价格、是否现金支付、一次共同出售的奴隶人数），以及买方和卖方的信息。Fogel 和 Engerman 利用这些数据分析州际奴隶贸易，并讨论奴隶经济在经济上的可持续性问题（Fogel and Engerman，1974）。

Figure 14.2 Facsimile of 1950 Census Schedule for Orange City, lowa (right side of page)
图 14.2 1950 年爱荷华州奥兰治城人口普查表摹本（页面右侧）

Original figure / 原图

The uploaded Markdown contains this figure as a remote image link.
上传的 Markdown 中该图以远程图片链接形式存在。

Open original image / 打开原图

The work of building the corpus of machine-readable databases began in the 1960s, and continues both with small compilations and large collaborative data projects. In addition to the IPUMS project mentioned above, one can find large-scale historical data compilations of cost-of-living studies, election results, crime data, and the records of the heights and weights of people in the past. The creation and retrieval of historical data has also led to revision and improvement of data series compiled in earlier years and to the analysis of the history of data development. Most recently, for example, economic historians have produced a new ‘millennial’ edition of the Historical Statistics of the United States (Carter et al., 2006), which promises to provide opportunities for even more quantitative historical analysis.

构建机器可读数据库资料库的工作始于 20 世纪 60 年代，并一直延续至今，既包括小型汇编，也包括大型合作数据项目。除前文提到的 IPUMS 项目外，还可以看到关于生活成本研究、选举结果、犯罪数据，以及历史人口身高和体重记录等大规模历史数据汇编。历史数据的建立与重新获取，也促使早年编制的数据序列得到修订和改进，并推动了对数据发展史本身的分析。最近，例如经济史学家编制了新版“千年版”《Historical Statistics of the United States》（Carter et al.，2006），预计将为更广泛的量化历史分析提供新的机会。

Figure 14.3 Slave Bill of Sale, Davidson County, Tennessee, 1833
图 14.3 1833 年田纳西州戴维森县奴隶买卖契约

Original figure / 原图

The uploaded Markdown contains this figure as a remote image link.
上传的 Markdown 中该图以远程图片链接形式存在。

Open original image / 打开原图

Analyzing Historical Data Sets
分析历史数据集

Sampling and the Universe of Cases
抽样与案例总体

As have data analysts in the other social sciences, historians have made use of the theory of probability sampling to reduce the volume of information for a particular study to a manageable level. Just as one does not need to survey the entire electorate to develop quite precise estimates of the ultimate election results, so historians studying family structure or economic activity or consumer behavior have not had to record all such behavior for study. As noted above, the process of creating historical data sets is sufficiently time-consuming to strongly recommend sampling strategies designed to reduce the volume of coding and data entry to the minimum necessary for robust analysis. Thus the original users and secondary users of the archived historical data sets need to attend to sampling strategy and introduce appropriate sample weights and measures of error into the analysis.

与其他社会科学领域的数据分析者一样，历史学家也利用概率抽样理论，把某项研究所需处理的信息量降低到可管理的程度。正如为了相当精确地估计最终选举结果并不需要调查所有选民一样，研究家庭结构、经济活动或消费者行为的历史学家也不必把所有相关行为全部记录下来。正如前文所述，建立历史数据集极其耗时，因此非常有必要采用抽样策略，把编码和数据录入的数量压缩到获得稳健分析所必需的最低程度。由此，档案历史数据集的原始使用者和二次使用者都必须关注抽样策略，并在分析中引入适当的样本权重和误差度量。

A more difficult issue is the one facing the historian who cannot be sure that she knows what the universe of cases actually is. Do the extant newspaper reports of lynchings, for example, encompass all lynchings (Griffin et al., 1997; Tolnay and Beck, 1995)? Are the records of wills filed with a particular county complete, or might some have been destroyed or lost over the centuries? These dilemmas have their analogues in non-quantitative research. But as with code-book creation, the research must provide best estimates of answers to such questions before analysis, and a substantial methodological literature has emerged to address the issues, often with specific reference to the kind of data set being compiled.

更困难的问题，是历史学家无法确定自己是否真正知道案例总体究竟是什么。例如，现存报纸关于私刑的报道是否涵盖了所有私刑事件（Griffin et al.，1997；Tolnay and Beck，1995）？某个县登记的遗嘱记录是否完整，还是其中一些可能在几个世纪中被毁坏或遗失？这些困境在非量化研究中也有对应形式。不过，与建立编码手册一样，研究者在分析之前必须对这类问题给出最佳估计；围绕这些问题已经形成了相当丰富的方法论文献，而且往往会针对正在编制的数据集类型提出具体讨论。

Techniques of Analysis
分析技术

Statistical analysis of historical data has ranged from elementary data analysis of the patterns of central tendency and dispersion of the phenomena under study to elaborate explanatory models of events and behavior. Much historical quantitative analysis has been descriptive, simply excavating and documenting patterns of change and activity in quantitative form that cannot be revealed by traditional historical analysis. Thus much work—important work—is simple counting of a phenomenon, and describing trends over time.

历史数据的统计分析，既包括对研究现象的集中趋势和离散程度进行基础分析，也包括建立复杂的事件与行为解释模型。大量量化历史分析本质上是描述性的，其工作只是以量化形式发掘并记录传统历史分析无法揭示的变

化与活动模式。因此，很多工作——而且是重要的工作——就是简单地计数某一现象，并描述其随时间变化的趋势。

Somewhat more elaborate analysis involves determining the correlates of the phenomenon under study, or building a model to explicate more complex patterns in the data. Here the standard bivariate and multivariate techniques of statistics provide the tools necessary for the analysis. Quantitative historians have borrowed heavily from sociology, political science, demography and economics, and made use of the classic linear regression model and its variants as the workhorse technique for more complex analysis. Statistical packages, such as SPSS, SAS, STATA and the like underpin the analysis of quantitative historical work, as they do for the social sciences.

更复杂一些的分析，会考察研究现象与其他因素之间的相关关系，或者建立模型来说明数据中更复杂的模式。此时，标准的双变量和多变量统计技术提供了必要的分析工具。量化史学家大量借鉴社会学、政治学、人口学和经济学，并把经典线性回归模型及其各种变体作为复杂分析的核心技术。SPSS、SAS、STATA 等统计软件和类似工具，正如在社会科学中一样，也构成量化历史研究分析工作的基础。

There is some evidence that quantitative history has begun to have an impact on the larger methodological practice of the social sciences, as quantitative historians have brought their methodological expertise to the social sciences. Two brief examples should illustrate that impact.

有一些证据表明，量化史学已经开始影响社会科学更广泛的方法论实践，因为量化史学家把自己的方法专长带回了社会科学。下面两个简短例子可以说明这种影响。

The first is development of the field of ecological regression, particularly for analysis of electoral patterns. Political scientists can supplement analysis of election results with surveys of individual voters. Indeed the National Election Survey, conducted since 1948, has itself become an historical source of changing electoral behavior. But historians cannot go back and survey voters from the election of 1860, and thus must make use of the aggregate election results and the ecological characteristics of the voting units—e.g., precincts, districts or counties—that provided the vote. Ecological inference suffers from the threat of the ecological fallacy, that is, the danger of wrongly inferring individual level behavior from the patterns of aggregates. Practitioners of quantitative history have taken up new methods developed by political scientists and have devoted good effort to minimizing, if not completely solving, this dilemma. With historically minded political scientists, they have produced a methodological literature and new techniques that have produced rigorous results.14

第一个例子是生态回归领域的发展，尤其是其在选举模式分析中的应用。政治学家可以用对个体选民的调查补充选举结果分析。事实上，自 1948 年开始实施的 National Election Survey 本身已经成为研究选举行为变化的一项历史资料。但是，历史学家不可能回到过去调查 1860 年选举的选民，因此必须使用汇总的选举结果，以及产生这些选票的投票单位——例如选区、地区或县——的生态特征。生态推断面临“生态谬误”的风险，也就是错误地从总体层级的模式推断个体层级行为。量化史学研究者采用政治学家发展出的新方法，并投入大量努力来尽量减少、即使尚不能完全消除这一难题。他们与具有历史意识的政治学家一道形成了一套方法论文献，并发展了能够产生严格结果的新技术。[14]

The second contribution is serious attention to the development of statistical techniques to conceptualize and model time and temporal explanations. The methodological bread and butter for all historians is ‘thinking in time’ (Neustadt and May, 1986), and that standpoint has prompted historians and historically attuned social scientists to think about how to develop techniques of statistical analysis suitable for the goals of historical analysis.

第二项贡献，是认真推动用于概念化并建模时间与时间性解释的统计技术。对于所有历史学家而言，“在时间中思考”（thinking in time）（Neustadt and May，1986）都是方法论上的基本功；这一立场促使历史学家以及具有历史意识的社会科学家思考，如何发展适合历史分析目标的统计分析技术。

Historians think about questions of what is an event, how is it bounded and measured; what is a turning point; what is a transition; what is a conjuncture or a rupture; and how is a period of time organized and bounded. Economists and other social and biological scientists have developed techniques to measure time series and temporal and cyclical events, for example, life cycles. The entry of quantitative historians into these discourses has been a useful clarification of the methodological issues involved. For example, the phrase, ‘longitudinal analysis’ that social scientists use does not necessarily privilege time as a central concern for analysis. Historians and social scientists who make temporal analysis such a central concern have thus argued for the need to add methods that will address ‘thinking in time’ to the standard repertoire of statistical techniques. Such techniques as sequence analysis, event history analysis and the methodological discussions surrounding autocorrelation in time series analysis have usefully been enriched by the growth of the field of quantitative history.15

历史学家会思考：什么是一个事件，它如何被界定和测量；什么是转折点；什么是转型；什么是合流（conjuncture）或断裂（rupture）；一个时期又如何被组织和划定边界。经济学家以及其他社会科学和生物科学研究者已经发展出测量时间序列和时间性、周期性事件（例如生命周期）的技术。量化史学家进入这些讨论，有助于进一步澄清其中涉及的方法论问题。例如，社会科学家常用的“纵向分析”（longitudinal analysis）一词，并不必然把时间置于分析的核心。正因如此，那些把时间分析置于中心的历史学家和社会科学家主张，应当在标准统计技术体系中加入能够真正处理“在时间中思考”的方法。序列分析、事件史分析，以及时间序列分析中围绕自相关问题展开的方法论讨论，都因量化史学的发展而得到有益推进。[15]

The Costs of Doing Quantitative History

开展量化史学研究的成本

The cost of scholarly work in quantitative history, like the cost of all scholarly work, can be measured in terms of both time and money required for the scholarship to flourish. The largest change in the working environment since the 1960s is that computing costs, which were quite expensive in the early years of the field, have dropped as the larger information revolution has developed. To my knowledge, there is no extant scholarly analysis of the costs of quantitative history versus traditional history, though I suspect that the underlying funding situation for quantitative historians has had an effect on the progress of the field.

量化史学研究与所有学术研究一样，其成本既可以用研究得以开展所需的时间衡量，也可以用金钱衡量。自 20 世纪 60 年代以来，研究工作环境中最大的变化是计算成本下降：在该领域早期，计算费用相当高昂，但随着更广泛的信息革命发展，这些成本不断降低。据我所知，目前并不存在对量化史学与传统史学研究成本进行系统比较的学术分析，不过我怀疑，量化史学家的基础经费状况确实影响了这一领域的发展进程。

In the early years of the development of quantitative history, in the United States the Social Science Research Council, the American Historical Association, the National Endowment for the Humanities, and the National Science Foundation, as well as research universities around the country, all provided sponsorship of the field by funding grants for data development, conference sponsorship and the institutional work required to promote the field. This early institutional support was aimed at jump-starting the field, not at providing sustained long-term support. Related to this, the National Endowment for the Humanities, the main federally sponsored grant agency for historians, has a much lower funding level than the federal funding agencies that support related social science research—for example, the National Science Foundation or the National Institutes of Health. The United States, unlike European nations, does

not include history in the main governmental foundation for funding academic research. Accordingly, quantitative history projects in the United States have had major difficulty in competing with both large-scale humanities grant projects, such as compilations of archival papers, and with large-scale long-term research projects in the social sciences such as the National Election Study or the Panel Study of Income Dynamics. Allan Bogue (1983) identified the chronic problems of funding faced by quantitative historians in the late 1970s. They remain unsolved as the concrete example which follows illustrates.

在量化史学发展的早期，美国社会科学研究委员会、美国历史学会、美国国家人文基金会和美国国家科学基金会，以及全国各地的研究型大学，都通过资助数据建设、会议组织和推动这一领域所需的制度性工作，为量化史学提供支持。早期的制度支持旨在帮助这一领域启动，而不是长期持续供给。与此相关的是，作为历史学家主要联邦资助机构的美国国家人文基金会，其经费规模远低于支持相关社会科学研究的联邦资助机构，例如美国国家科学基金会或美国国立卫生研究院。与欧洲国家不同，美国没有把历史学纳入政府主要的学术研究资助体系。因此，美国的量化史学项目既难以与大型人文学科项目（如档案文献汇编）竞争，也难以与社会科学中的大型长期研究项目（如 National Election Study 或 Panel Study of Income Dynamics）竞争经费。Allan Bogue（1983）指出了 20 世纪 70 年代末量化史学家长期面临的经费问题。正如下一个具体案例所示，这些问题至今仍未得到解决。

Robert Fogel, by any measure, represents one of the most successful and innovative quantitative history scholars in the field, yet even he has faced major funding obstacles. Fogel was awarded the Nobel Prize in Economics in 1993, and in his autobiographical statement prepared for the award, he described his career and acknowledged the problems of funding he faced, particularly, as he put it, for the ‘current research projects on which I reported in the Prize Lecture’. The Center for Population Economics at the University of Chicago and the Walgreen Chair provided funding when federal grants would not. ‘The data on health conditions’, he wrote\:comes from a project called ‘Early Indicators of Later Work Levels, Disease, & Death’ which is tracing nearly 40,000 Union Army men from the cradle to the grave. It takes over 15,000 variables to describe the life-cycle history of one of these men. These life-cycle histories are created by linking about a score of data sets. It took more than half a decade of work to investigate the potential of these data sets, work out procedures for data retrieval and file management, and to establish the feasibility of the enterprise in our own minds.The site committee of the National Institutes of Health which reviewed the original project proposal in 1986 agreed that such a project could in principle make a significant contribution to an understanding of the process of aging, but they were skeptical about the quality of some of the data, about whether the software and programming procedures we had developed by that time were adequate for the management of such a large data set, and about whether the project could be completed within the proposed budget.To resolve these doubts it was necessary to draw a six percent subsample which linked together all of the separate sources and which demonstrated the effectiveness of the software by analyzing the information in the subsample. It took an additional four years to complete the second phase of the justification of the project. Thus nearly a decade of preliminary research, much of it funded by Walgreen and the CPE, was required before the project was accepted by the peer reviewers of NIH and NSF.16 (Fogel, 1993)

无论按何种标准衡量，Robert Fogel 都是这一领域最成功、最具创新性的量化史学学者之一，但即便如此，他也曾面临重大的经费障碍。Fogel 于 1993 年获得诺贝尔经济学奖，他在为获奖撰写的自传性陈述中回顾自己的学术生涯，并承认曾面临资助困难，尤其是他所谓“我在诺贝尔奖演讲中报告的当前研究项目”。当联邦拨款无法提供支持时，芝加哥大学人口经济学中心和 Walgreen Chair 提供了经费。他写道：“有关健康状况的数据来自一个名为‘Early Indicators of Later Work Levels, Disease, & Death’的项目，该项目追踪近 4 万名联邦军士兵从出生到死亡的一生。要描述其中一名士兵的生命周期史，需要超过 1.5 万个变量。这些生命周期史通过连接大

约二十个数据集建立起来。我们花了五年多时间研究这些数据集的潜力，制定数据提取和文件管理程序，并首先在我们自己看来证明这一项目是可行的。1986 年，美国国立卫生研究院审查最初项目申请的现场评审委员会同意，这样的项目原则上可以为理解衰老过程作出重要贡献，但他们对部分数据的质量有所怀疑，也怀疑我们当时开发的软件和编程程序是否足以管理如此庞大的数据集，以及项目能否在拟定预算内完成。为消除这些疑虑，我们必须抽取一个 6%的子样本，把所有分散的资料来源连接起来，并通过分析这一子样本的信息证明软件的有效性。完成项目论证的第二阶段又花了四年。因此，在项目最终获得 NIH 和 NSF 同行评审者认可之前，我们进行了近十年的前期研究，其中很大一部分由 Walgreen 和 CPE 资助。”[16]（Fogel，1993）

Despite such barriers, quantitative historians have been able to take advantage of the technological developments in computing and data management to make major advances in the ease of analysis, in terms of both time and money. For example, historians of the 1960s through the 1980s who wished to have access to the archived data sets at ICPSR had to order tapes and paper codebooks which were delivered by mail. The tape was then mounted on a mainframe computer, to be accessed in a statistical package run in a mainframe environment (with computer usage often charged by the university in the same way that phones or paper were charged). By the early 1990s, users could access files using FTP (file transfer protocol), and micro-computers on university desktops were providing direct access to statistical packages, even if those programs were sometimes still lodged on a mainframe. By the mid-1990s, desktop computing had replaced mainframe computing for most applications, and by the early 2000s, ICPSR initiated ICPSR Direct, the application that permitted an authorized user to download data files and PDF code-books directly to a desktop.

尽管存在这些障碍，量化史学家仍能利用计算技术和数据管理的发展，大幅提高分析在时间和金钱成本上的便利程度。例如，20 世纪 60 至 80 年代，希望使用 ICPSR 存档数据集的历史学家必须订购磁带和纸质编码手册，并等待邮寄送达。随后，磁带需要装载到大型主机上，再通过运行于大型主机环境中的统计软件包访问（大学往往像收取电话费或纸张费用一样，按计算机使用量收费）。到 20 世纪 90 年代初，使用者可以通过 FTP（文件传输协议）访问文件，大学桌面上的微型计算机也开始直接连接统计软件包，尽管这些程序有时仍实际运行在大型主机上。到 20 世纪 90 年代中期，桌面计算机在多数应用中已取代大型主机；进入 21 世纪初，ICPSR 推出 ICPSR Direct，使获授权的使用者能够把数据文件和 PDF 编码手册直接下载到桌面计算机。

Critiques of Quantitative History
对量化史学的批评

From the outset of the development of the field of quantitative history, powerful critics have challenged practitioners on their work, and even challenged the usefulness of the field itself. In the early 1960s, Carl Bridenbaugh devoted a portion of his 1962 American Historical Association Presidential Address to a condemnation of quantitative history (Bridenbaugh, 1962), memorably labeling it a ‘bitch goddess’ (Bogue, 1983). Even during the period of the rapid growth of quantitative history in the 1960s and 1970s, ‘traditional’ historians expressed doubts about the new methods, challenging them as reductionist, brittle and not pertinent to the main goal of the historical narrative. Critics were extremely dubious of the ‘scientific’ claims of quantitative historians, and resisted the challenge of the quantifiers that traditional historical writing was not theoretically rigorous or conceptually consistent.

从量化史学领域形成之初起，就有有影响力的批评者质疑这一领域研究者的工作，甚至质疑该领域本身是否有用。20 世纪 60 年代初，Carl Bridenbaugh 在 1962 年的美国历史学会主席演说中专门用一部分篇幅批评量化史学（Bridenbaugh，1962），并令人印象深刻地把它称为“bitch goddess”（Bogue，1983）。即便在 20 世纪 60 至 70 年代量化史学迅速发展的时期，“传统”历史学家仍对这些新方法表示怀疑，批评它们具有还原主义

倾向、过于僵硬，并且与历史叙事的主要目标无关。批评者对量化史学家所谓“科学性”的主张极为怀疑，也拒绝接受量化研究者对传统历史写作提出的挑战——即传统历史写作在理论上不够严格、在概念上不够一致。

In the 1980s, some of the original proponents of the field also renounced their earlier enthusiasm and suggested that quantitative methods had not fulfilled their promise. Most notable among these critics were Lee Benson and Lawrence Stone, early enthusiasts who had changed their minds (Benson, 1984; Stone, 1977, 1979). Such recantations gave support to the anti-quantifiers at a time when major new methodological challenges were facing historians, most notably from the postmodernists and what came to be called ‘the cultural turn’. Through this welter of debate, quantitative practitioners continued their efforts, somewhat chastened by their fall from the heights of fashion of earlier years, but grounded sufficiently institutionally and intellectually to continue to work.17

到 20 世纪 80 年代，这一领域最初的一些倡导者也放弃了早期的热情，并提出量化方法并没有兑现其承诺。其中最著名的批评者包括 Lee Benson 和 Lawrence Stone，这两位早期的支持者后来改变了看法（Benson，1984；Stone，1977，1979）。在历史学正面对新的重大方法论挑战、尤其是后现代主义以及后来被称为“文化转向”的潮流时，这些转变为反量化立场提供了支持。在纷繁复杂的争论中，量化研究者仍继续推进工作；尽管他们已经从早年最流行的位置跌落下来并受到一定挫折，但其制度与知识基础仍足以支撑这一领域继续发展。[17]

Through some twenty years of debate, neither side of the traditional/quantitative divide ‘won’ their arguments. Rather, by the 1990s, the debate cooled into something of an uneasy truce, with practitioners acknowledging some of the points of their opponents, but agreeing to disagree on the larger validity of their enterprise.18 In practical terms, quantitative techniques did not become a routine part of history graduate student training as they did in the social sciences, but have remained a specialty of some historians in some graduate training programs, considered more akin to language requirements for reading historical literature and texts of a non-English-speaking society than to a methodological necessity for all practicing historians. This compartmentalization of the skills of quantification for historians has in turn affected the practice of quantitative historians within the larger history profession.

经过大约二十年的争论，传统史学与量化史学之争并没有任何一方真正“获胜”。相反，到 20 世纪 90 年代，这场争论逐渐冷却为一种并不安稳的休战：双方研究者承认对方的部分观点，但在各自事业的整体有效性上仍然保留分歧。[18] 在实践层面，量化技术并没有像在社会科学中那样成为历史学研究生训练的常规组成部分，而是继续作为部分历史学家、部分研究生项目中的专门技能。人们往往把它看得更像阅读非英语社会的历史文献和文本所需的语言要求，而不是所有历史学实践者都必须掌握的方法论能力。量化技能在历史学家中的这种分区化，也进一步影响了量化史学家在更广泛历史学专业内部的研究实践。

History as a field has maintained its roots as a ‘humanities’ discipline and quantitative historians’ connections to the social sciences seem to many to be a betrayal of the historical project. The methodological ‘training gap’ has meant that when quantitative historians research and write for other historians, as opposed to other social scientists, they cannot expect their readers to appreciate or even understand the technical issues involved in their work. The history profession has maintained its commitment to accessible writing as well, and thus when writing for the broader audience of historians, quantitative historians have had to avoid technical jargon—for example, by avoiding the use of variable names in the explication of a model—and be mindful to explicate their arguments clearly.

历史学作为一个学科一直保持着其“人文学科”的根基，因此在许多人看来，量化史学家与社会科学的联系似乎背离了历史研究本身。这种方法论上的“训练鸿沟”意味着，当量化史学家的研究和写作对象是其他历史学家而非其他社会科学家时，他们不能期待读者能够欣赏、甚至理解其工作中的技术问题。与此同时，历史学界一直

重视可读性，因此面向更广泛的历史学读者写作时，量化史学家不得不避免技术行话——例如在解释模型时避免直接使用变量名称——并注意清楚说明自己的论证。

The critiques have also encouraged quantitative historians to attend to the limitations in statistical methodology for analyzing historical processes, as discussed above. Much of this new work on statistical techniques for analyzing temporal processes is still in development and has yet to provide enough empirical work to demonstrate the robust nature of the new techniques, and hence convince non-quantitative historians, as well as the larger social science community, of the need to integrate explicitly temporal analysis into basic methods. But the promise is there, and as noted below, there are encouraging signs on the horizon.

这些批评也促使量化史学家更加关注统计方法在分析历史过程时的局限，如前文所述。关于如何用统计技术分析时间过程的许多新工作仍在发展之中，目前还没有足够多的实证研究能够证明这些新技术具有稳健性，从而说服非量化历史学家以及更广泛的社会科学界，把明确的时间分析纳入基本方法体系。不过，这些方法确实具有潜力；如下文所示，也已经出现了一些令人鼓舞的迹象。

The Future of the Field

这一领域的未来

The intellectual achievements of quantitative history in conjunction with the larger information technology revolution make the prognosis for the future of the field better today than it has been for many years.19Almost a half-century on, one can look back at steady development, though not always in a satisfyingly linear pattern.20 Perhaps the most interesting recent development is the impact of the information technology revolution on the larger practice of historians. When quantitative history as a field was in its most rapid initial development, most traditional historians labored much as their nineteenth-century predecessors had with pen, pencil, typewriter and note-card as technological support. Bibliographic work entailed using library card catalogs or reading large indexed tomes of articles, books, compilations, and the like. ‘Data management’ meant developing a file of index cards, not an electronic spread sheet or database. Secretaries typed manuscripts for publication, and though some large research institutions had introduced line editors for manuscript production by the 1970s, these were machines for staff, not faculty or students. By the 1980s, the situation changed. Desktop computers proliferated and for most historians, word processing opened up the possibility of the electronic future. By the 1990s, email replaced typed letters. After 1995, the content on the internet exploded, and first bibliographical work, and then much actual archival work, shifted to a computerized format. In short, non-quantitative historians had come to operate in a technological environment that was very similar to their quantifying peers. Most recently, cheap computing has made multimedia evidence—visual and oral, video and audio—accessible to the practicing historian. One can see these developments in particularly acute form in the developing field of historical geographic information systems, or historical GIS. GIS, was until quite recently, a very expensive technology, and thus adding historical maps to geographic databases has only just begun. As with the digitizing projects of the 1960s and 1970s, the payoff for the large initial costs of first translating maps to a new medium to become ‘data’, and then the development of new theory, software programs and methods to make the best use of these new data, are just beginning (Knowles, 2006).

量化史学的知识成就与更广泛的信息技术革命相结合，使今天这一领域的未来前景比过去许多年都更好。[19]近半个世纪之后回看，我们可以看到它一直在稳步发展，尽管这种发展并不总是呈现令人满意的线性轨迹。[20]近期最有趣的发展，也许是信息技术革命对历史学家整体研究实践的影响。在量化史学最初快速发展的时期，大多数传统历史学家的工作方式与 19 世纪的前辈差别并不大，主要依靠钢笔、铅笔、打字机和卡片作为技术辅助。书目工作意味着查阅图书馆卡片目录，或翻阅带有大型索引的论文、书籍和汇编。“数据管理”指的是建立索引

卡片档案，而不是电子表格或数据库。秘书负责把论文打字成稿；尽管到 20 世纪 70 年代，一些大型研究机构已经引入行式编辑器来处理稿件，但这些机器是工作人员使用的，而不是教师或学生使用的。20 世纪 80 年代，情况发生变化。桌面计算机快速普及，文字处理软件为大多数历史学家打开了电子化未来的可能。到 20 世纪 90 年代，电子邮件取代了打字信件。1995 年以后，互联网内容爆炸式增长，先是书目工作，随后许多真正的档案研究也转向计算机化形式。简言之，非量化历史学家也开始在一种与量化研究同行非常相似的技术环境中工作。最近，廉价计算又使多媒体证据——视觉与口述材料、视频与音频——进入普通历史学家的研究实践。在正在发展的历史地理信息系统（historical GIS）领域，这些变化表现得尤其明显。直到不久前，GIS 仍是一项非常昂贵的技术，因此把历史地图加入地理数据库的工作才刚刚起步。正如 20 世纪 60 至 70 年代的数字化项目一样，最初需要付出高昂成本，把地图转换到新媒介中使其成为“数据”，随后还要发展新的理论、软件和方法来充分利用这些新数据；这些投入的回报如今才刚刚开始显现（Knowles，2006）。

More broadly, the effect of these technological changes has been to produce a convergence of work of what one might call ‘technologically enabled’ history. Traditional historians and humanists in general—for example, in the work of Franco Moretti (2005)—also now work with electronic databases, learn new computer programs to analyze the rapidly proliferating data, and explore new forms of presentation of the results of their analysis. Quantitative historians had to learn the skills necessary to prepare and present statistical results in print. Historians more generally are using visual images, audio and video in their presentations, not as ‘illustration’ to enhance or supplement an analysis but as core evidence for analysis.21

从更广泛的层面看，这些技术变化产生的影响，是促成了一种可以称为“技术赋能的历史学”（technologically enabled history）的研究汇流。传统历史学家以及更广泛的人文学者——例如 Franco Moretti（2005）的研究——现在也会使用电子数据库，学习新的计算机程序来分析迅速扩张的数据，并探索新的方式来呈现分析结果。量化史学家过去必须学习如何在印刷出版物中准备并展示统计结果；如今，更广泛的历史学家也在展示中使用视觉图像、音频和视频，而且不再把它们仅仅当作增强或补充分析的“插图”，而是把它们作为分析的核心证据。[21]

Richard Steckel (2007) recently proposed an agenda for what he called ‘Big Social Science History’, which would extend the capacities of quantitative history and translate some of its methods of work to non-quantitative projects.22 Andrew Abbott (2005) has also proposed such possibilities. As with the first generation of quantitative history, these large agendas will require collaborative efforts to manage the enormously expanding data infrastructure and the myriad computer technologies required to make best use of the expanding corpus of digitized historical evidence, and to develop appropriate theoretical approaches to such historical work.

Richard Steckel（2007）近期提出了一个他称为“大社会科学史学”（Big Social Science History）的研究议程，希望扩展量化史学的能力，并把其中一些工作方法转用于非量化项目。[22] Andrew Abbott（2005）也提出了类似可能。和第一代量化史学一样，这些宏大的研究议程需要通过合作来管理快速膨胀的数据基础设施，以及为充分利用不断增长的数字化历史证据所需的众多计算机技术，同时还要为这类历史研究发展适当的理论路径。

Acknowledgements
致谢

*This essay has been improved considerably by comments from colleagues, particularly Peggy Adams, Erik Austin, Morgan Kousser, Jim Oberly, Lex Renda, Jack Reynolds, Steve Ruggles, Carole Shammas and Dan Scott Smith, The overall interpretation and remaining errors are all mine.

本文因同事们的意见而得到显著改进，尤其感谢 Peggy Adams、Erik Austin、Morgan Kousser、Jim Oberly、Lex Renda、Jack Reynolds、Steve Ruggles、Carole Shammas 和 Dan Scott Smith。本文的整体解释以及仍然存在的错误均由作者本人负责。